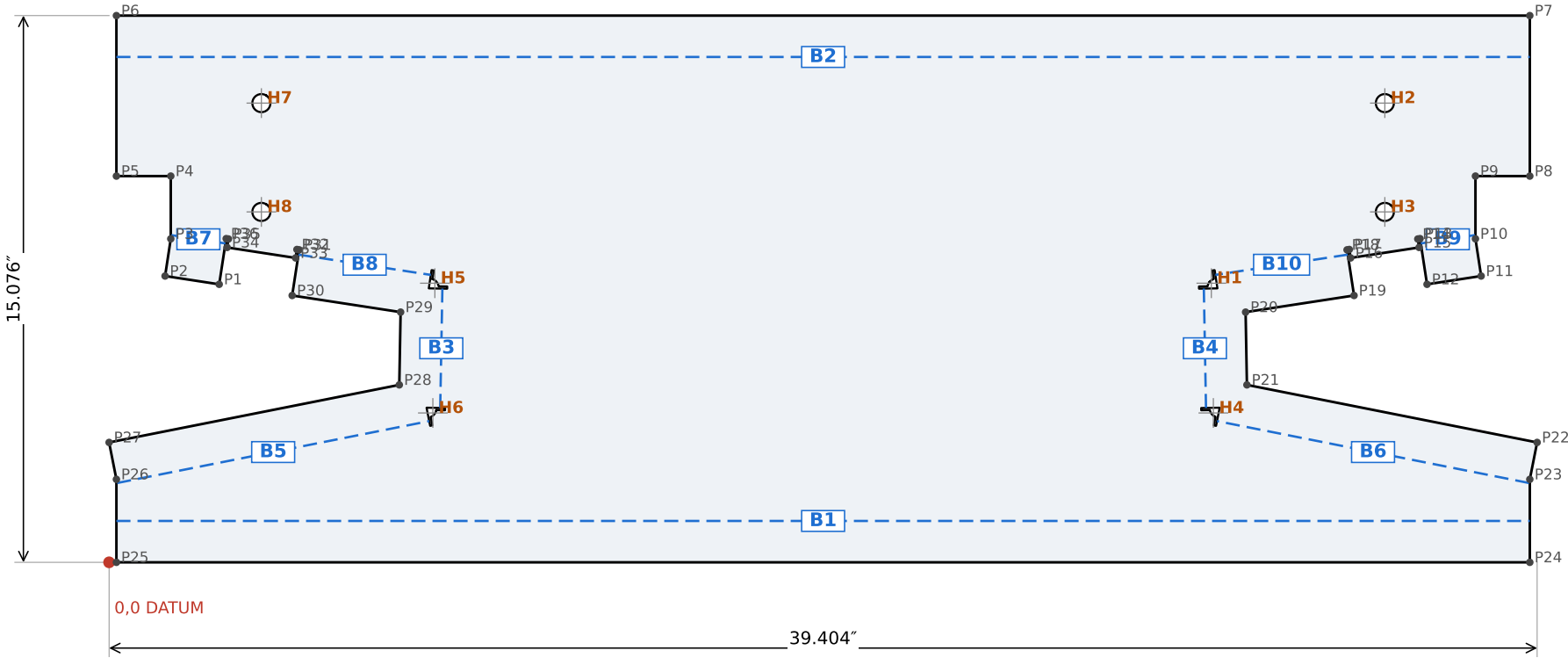


FF-3 — blank 1 of 4

FF3-1 top plate — flat pattern, 10 bends 90° down · blank 39.404” × 15.076” · HR steel sheet 0.118” · 1 per vehicle



HOLES (centre X, Y from datum)

H1	slot	0.509 × 0.505"	@1°	X	30.414	Y	7.696
H2	Ø0.500"			X	35.202	Y	12.663
H3	Ø0.500"			X	35.202	Y	9.663
H4	slot	0.511 × 0.499"	@1°	X	30.470	Y	4.118
H5	slot	0.509 × 0.505"	@-1°	X	8.989	Y	7.696
H6	slot	0.511 × 0.499"	@-1°	X	8.934	Y	4.118
H7	Ø0.500"			X	4.202	Y	12.663
H8	Ø0.500"			X	4.202	Y	9.663

BENDS (bend centre line, from datum) IR = t = 0.118"

B1	(3)	90° ↓	(0.202, 1.141) → (39.202, 1.141)
B2	(4)	90° ↓	(0.202, 13.935) → (39.202, 13.935)
B3	(5)	90° ↓	(9.137, 4.258) → (9.196, 7.547)
B4	(6)	90° ↓	(30.267, 4.258) → (30.207, 7.547)
B5	(7)	90° ↓	(0.222, 2.187) → (8.833, 3.894)
B6	(8)	90° ↓	(39.182, 2.187) → (30.571, 3.894)
B7	(9)	90° ↓	(1.717, 9.022) → (3.210, 8.793)
B8	(10)	90° ↓	(5.224, 8.485) → (8.878, 7.925)
B9	(11)	90° ↓	(37.686, 9.022) → (36.193, 8.793)
B10	(12)	90° ↓	(34.179, 8.485) → (30.526, 7.925)

OUTLINE POINTS (X, Y from datum)

P1	3.038	7.666
P2	1.545	7.895
P3	1.702	8.921
P4	1.702	10.651
P5	0.202	10.651
P6	0.202	15.076
P7	39.202	15.076
P8	39.202	10.651
P9	37.702	10.651
P10	37.702	8.921
P11	37.859	7.895
P12	36.366	7.666
P13	36.173	8.925
P14	36.111	8.916
P15	36.147	8.683
P16	34.257	8.393
P17	34.221	8.627
P18	34.159	8.617
P19	34.352	7.358
P20	31.360	6.899
P21	31.396	4.893
P22	39.404	3.306
P23	39.202	2.288
P24	39.202	0.000
P25	0.202	0.000
P26	0.202	2.288
P27	0.000	3.306
P28	8.008	4.893
P29	8.044	6.899
P30	5.052	7.358
P31	5.245	8.617
P32	5.183	8.627
P33	5.147	8.393
P34	3.257	8.683
P35	3.292	8.916
P36	3.231	8.925

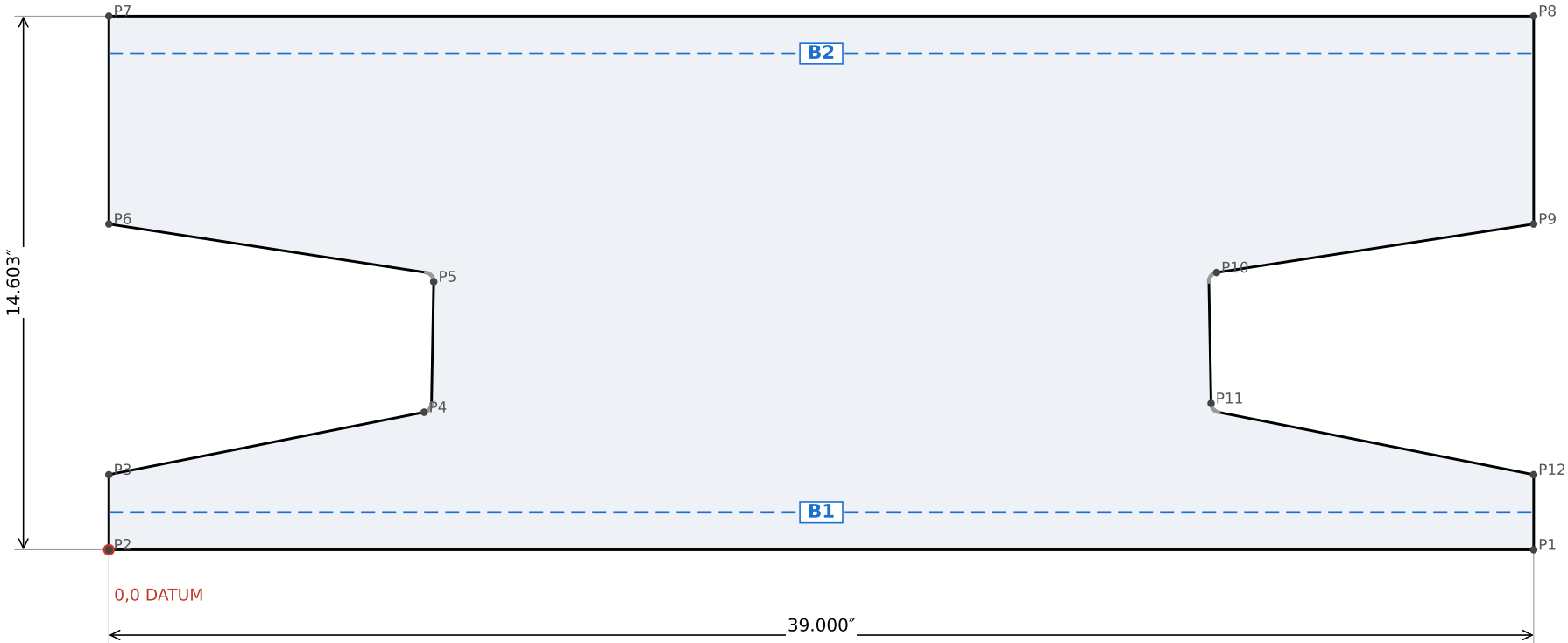
NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 2. Material: HR steel sheet 0.118" (t). Inside bend radius = t.
 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * rims 1½" bent down on the long sides and around the whole notch; leg ends open; in the 4 inner notch corners the rims are mitred — a gap stays in the corner; 1/16" reliefs at the bend ends; the 4 Ø½ holes are cut right away

FF-3 — blank 1 of 4	HR steel sheet 0.118"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FF3-1 top plate — flat pattern, 10 bends 90° down	QTY: 1 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 4

FF-3 — blank 2 of 4

FF3-2 lower plate — flat pattern, 2 bends 90° up · blank 39.000” × 14.603” · HR steel sheet 0.118” · 1 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 0.118” (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * rims on the long sides bent up 1” (to the FF3-1 underside), set in by t inside the FF3-1 rims; 1/16” reliefs at the bend ends

HOLES (centre X, Y from datum)

none

BENDS (bend centre line, from datum) IR = t = 0.118”

B1 (1) 90° (0.000, 1.022) → (39.000, 1.022)
B2 (2) 90° (0.000, 13.581) → (39.000, 13.581)

OUTLINE POINTS (X, Y from datum)

P1 39.000 0.000
P2 0.000 0.000
P3 0.000 2.052
P4 8.630 3.762
P5 8.892 7.333
P6 0.000 8.914
P7 0.000 14.603
P8 39.000 14.603
P9 39.000 8.914
P10 30.320 7.585
P11 30.168 4.003
P12 39.000 2.052

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

8.677 3.776
8.720 3.799
8.739 3.813
8.774 3.847
8.801 3.887
8.820 3.932
8.830 3.979
8.832 4.003

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

8.889 7.381
8.883 7.404
8.866 7.449
8.841 7.489
8.808 7.524
8.770 7.553
8.726 7.573
8.680 7.585

FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y

30.273 7.573
30.230 7.553
30.192 7.524
30.159 7.489
30.134 7.449
30.117 7.404
30.111 7.381
30.108 7.333

FREE-FORM CURVE 4 (grey dots, follow DXF) — points X, Y

30.170 3.979
30.180 3.932
30.199 3.887
30.226 3.847
30.261 3.813
30.280 3.799
30.323 3.776
30.369 3.762

FF-3 — blank 2 of 4	HR steel sheet 0.118”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FF3-2 lower plate — flat pattern, 2 bends 90° up	QTY: 1 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 2 / 4

FF-3 — blank 3 of 4

Passage wall 1⅛ × 4.69” (instead of the tube) · blank 4.694” × 1.125” · HR steel sheet 0.118” · 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	4.694	0.000
P2	4.694	1.125
P3	0.000	1.125
P4	0.000	0.000



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.118” (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3 — blank 3 of 4	HR steel sheet 0.118”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Passage wall 1⅛ × 4.69” (instead of the tube)	QTY: 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 3 / 4

FF-3 — blank 4 of 4

Passage wall 1⅓ × 5.02” (instead of the tube) · blank 5.021” × 1.125” · HR steel sheet 0.118” · 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	5.021	0.000
P2	5.021	1.125
P3	0.000	1.125
P4	0.000	0.000



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.118” (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3 — blank 4 of 4	HR steel sheet 0.118”	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Passage wall 1⅓ × 5.02” (instead of the tube)	QTY: 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 4 / 4